Homework

In this exercise, perform VaR backtesting for a portfolio that maintains a constant \$1 position in the S&P 500 index over time. Consider two different VaR estimation methods.

Parameters. Use $\alpha = 0.95$, $\beta = 0.05$, and $\lambda = \theta = 0.97$ in the EWMA specification.

Task 1. Download the daily closing prices of the S&P 500 index over the period 2016-05-23–2026-05-22. Present the first five and last five observations in the dataset, and report the total number of observations, denoted by N.

Task 2. Using the dataset above, produce a time series of one-day VaR estimates, based on the full loss operator, together with the corresponding exceedances under the empirical-distribution method. Use a four-year rolling window, so that the first VaR estimate is computed after the $M = 1010$th oldest log return.

Hint: For the portfolio with constant size 1 dollar, the full loss operator is

$$\ell_{[s]}(x) = -(e^x - 1).$$

Task 3. Repeat the VaR estimation on the same $J = N - M$ evaluation periods using the normal-log-returns model with EWMA updating. You will need an initial estimate of $\hat{\mu}$ and $\hat{\sigma}$ in order to obtain the first VaR estimate. It is acceptable to use the sample mean and sample variance of the first 1010 observations for this initialization.

Hint: With $l_{[t]}(x) = -e^x + 1$, if $X \sim N(\mu, \sigma^2)$, one can directly obtain its VaR at confidence level α by

$$\text{VaR} = 1 - e^{\mu + \sigma\Phi^{-1}(1-\alpha)}.$$

Task 4. Construct the $1 - \beta$ confidence interval for the number of exceedances and determine whether the observed number of exceedances falls within this interval.